

Artificial Intelligence (AI2002)

Date: 3/1/2024

Course Instructor(s)

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Roll No

Dr.

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Student Signature

Mr.

Mr.

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Sessional-I Exam

Total Time: 1 Hour

Total Marks: 12.5

Total Questions: 3

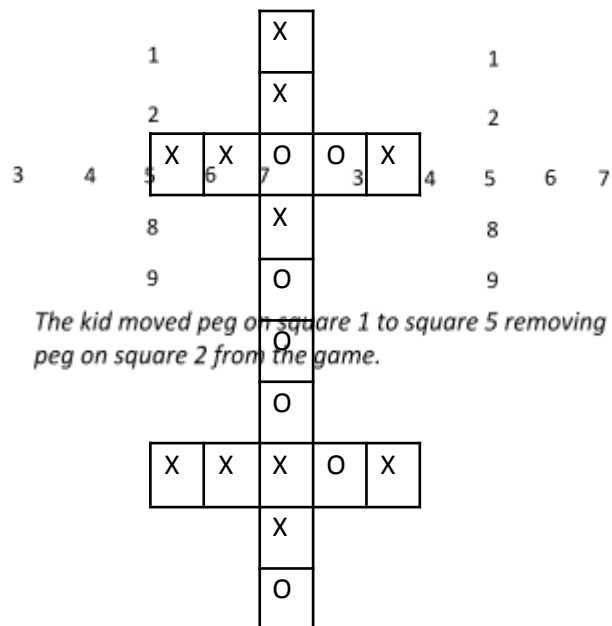
Semester: SP 2024

Campus: Karachi

Dept: School of Computing



Fig1: ORIGINAL 33 HOLES PEG SOLITAIRE



The kid moved peg on square 1 to square 5 removing the peg on square 2 from the game.

Fig2: KIDS 9 HOLE PEG SOLITAIRE: KID LEFT THE GAME IN BETWEEN

XS ARE PEGS AND OS ARE HOLES. KID'S LAST MOVE IS SHOWN

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- a. Consider a game of peg solitaire picture of actual game shown below. We will work on a toy peg solitaire (Fig2) game for kids with only 9 holes (instead of the original 33). Moreover, for kids we allow 2 pegs to be left anywhere on the board to win the game. Here are the rules for kids peg solitaire: -
- I. The object of the game is to remove all the pegs from the board except the last two which must finish anywhere in the game.
 - II. Commence the game by removing the center peg.
 - III. Players then take pieces by hopping over them into an empty space, removing from play the piece the player have just hopped over.
- Create a complete search tree starting from where the kid left the game and to the goal states where there are only two pegs left on the board. Label your nodes in the tree.
- b. You are making a FYP project “Intelligent medical drone” that helps people who need rescue during land sliding, or in any crowd panic situation. For making this project successful as a team leader you are selecting some features for your project. List task environment (PEAS) and types of environments. Identify which type of agent, apart from learning based, you will use to make your project work as an intelligent machine and why?

[1.5+2 = 3.5marks]

CLO 2: To identify and relate methods of blind as well as informed search.

Q2:

- a. In which scenario would bidirectional search perform the worst. Explain using examples.
- b. Consider the peg solitaire problem given above in question 1(a). Apply BFS and DFS strategy on the search tree and also list down the nodes coming in the frontier set and path returned by each strategy.

[1.5+3 = 4.5marks]

CLO 2: To identify and relate methods of blind as well as informed search.

Q3:

- a. For A* to return the optimal solution, the heuristic function used should be admissible and consistent. Explain what these two terms mean.
- b. Consider the peg solitaire problem given above in question 1(a). Apply A* search strategy on the search tree and also list down the nodes coming in the frontier set and path returned.

[1.5 + 3 = 4.5 marks]